

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	B Sathvika	Reg. No.:	192525224
Section / Batch:	2 nd year AIML	Date:	22/07/2026

Code of Conduct

I B Sathvika(192525224)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B Sathvika

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

QUESTION 1:

**SIMATS**
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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

4/10 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int main() {
3     int a[5] = {10,20,30,40,50};
4     int key, low = 0, high = 4, mid;
5     scanf("%d", &key);
6     while (low <= high)
7     {
8         mid = (low + high)/2;
9         if (a[mid] == key)
10        {
11            printf("found");
12            return 0;
13        }
14        else if (a[mid] < key)
15            low = mid + 1;
16        else
17            high = mid - 1;
18    }
19    printf("not found");
20    return 0;
21 }
```

Test Input
30

Output
found

QUESTION 2:

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[5];
3 int top = -1;
4 int pop()
5 {
6     if (top == -1)
7     {
8         printf("underflow");
9         return -1;
10    }
11    return stack[top--];
12 }
13 int main()
14 {
15     stack[++top] = 10;
16     stack[++top] = 20;
17     printf("%d", pop());
18     return 0;
19 }
20
```

Test Input
stdin input (optional)

Output
20

QUESTION 3:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

```
top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[5];
3 int top ;
4 void init()
5 {
6     top = -1;
7 }
8 int main() {
9     init();
10    return 0;
11 }
```

Test Input
-1

Output

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QUESTION 4:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <string.h>
3 char stack[100];
4 int top = -1;
5 void push(char c) {
6     stack[++top] = c;
7 }
8 char pop() {
9     return stack[top--];
10 }
11 int main() {
12     char infix[100], postfix[100];
13     int i, j=0;
14     char ch;
15     printf("enter infix expression:");
16 }
```

Test Input
stdin input (optional)

Output

Run Save Submit Fix

QUESTION 5:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 / Added a check to ensure top >= 0 so we don't read empty stack memory
2 if (top >= 0 && (ch == '[' && stack[top] == '[') || (ch == ']' && stack[top] == ']')) {
3     pop();
4 } else {
5     // Handle mismatched bracket or empty stack error here
6 }
```

Test Input
stdin input (optional)

Output
Compilation Error:
C:\Windows\TEMP\sandbox_192525224RB_6a959ade0c34e7.15783864\main.c
:1:1: error: expected identifier
or '(' before '/' token
1 / Added a check to ensure

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Submitted

QUESTION 6:



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BHAVANASI SATHVIKA
192525224RB

5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

temp = head
while temp.next != NULL:
 print(temp.data)
 temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 struct Node {
4     int data;
5     struct Node *next;
6 };
7
8 void printLinkedList(struct Node *head) {
9     struct Node *temp = head;
10
11     while (temp != NULL) {
12         printf("%d\n", temp->data);
13         temp = temp->next;
14     }
15 }
```

Test Input
5

Output
Compilation Error:
C:/CCPPComp/CCPP1/bin/./lib/gcc/x86_64-w64-mingw32/10.3.0/../../../../x86_64-w64-mingw32/bin/ld.exe:
C:/CCPPComp/CCPP1/bin/./lib/gcc/

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Submitted

QUESTION 7:



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BHAVANASI SATHVIKA
192525224RB

Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1#include <stdio.h>
2
3// Assuming stack is an array and top is passed by reference to update
4int POP(int stack[], int *top) {
5    if (*top == -1) {
6        printf("Underflow\n");
7        return -1; // Returning -1 to indicate error
8    } else {
9        int value = stack[*top];
10       (*top)--; // Correctly decrement the top index
11       return value;
12    }
13}
14
15int main() {
16    int stack[5] = {10, 20, 30};
17    int top = 2; // Points to index 2 (value 30)
18
19    int popped_value = POP(stack, &top);
20
21    if (popped_value != -1) {
22        printf
```

Test Input

stdin input (optional)

Output

QUESTION 8:



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BHAVANASI SATHVIKA
192525224RB

Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List

```
1 while (current != NULL) {
2     struct Node *next_node = current->next; // store next before changing
3     current->next = prev;
4     prev = current;
5     current = next_node;
6 }
```

Submitted

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192525224RB_6a959edff1bcb2.77385869\main.c
:1:1: error: expected identifier
or '(' before 'while'
1 | while (current != NULL) |

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QUESTION 9:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

Your Fixed Code

```
1 new->next = head;
2 head = new;
```

Submitted

Test Input

5

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192525224
RB_6a959f008d80f1.57867217\main.c
:1:4: error: expected '=', ',',
';', 'asm' or '__attribute__'
before '}' token
```

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QUESTION 10:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void DEQUEUE(int Q[], int *front, int *rear, int size) {
2     if (*front == -1) {
3         printf("Underflow\n");
4     } else {
5         int item = Q[*front];
6         printf("Dequeued: %d\n", item);
7
8         // Check if this was the last element in the queue
9         if (*front == *rear) {
10             *front = -1;
11             *rear = -1;
12         } else {
13             (*front)++;
14         }
15     }
16 }
```

Submitted

Test Input

5

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192525224
RB_6a959f4dc61561.54581272\main.c
: In function 'DEQUEUE':
C:\Windows\TEMP\sandbox_192525224
RB_6a959f4dc61561.54581272\main.c
```

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QUESTION 11:

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BHAVANASI SATHVIKA
192525224RB

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

10/10 answered

Your Fixed Code

```
1 while (expression[i] != '\0') {
2     if (isdigit(expression[i])) {
3         push(expression[i] - '0');
4     } else {
5         int op2 = pop(); // The first one popped is the right operand
6         int op1 = pop(); // The second one popped is the left operand
7         int result;
8
9         if (expression[i] == '+') result = op1 + op2;
10        else if (expression[i] == '-') result = op1 - op2;
11        else if (expression[i] == '*') result = op1 * op2;
12        else if (expression[i] == '/') result = op1 / op2;
13
14        push(result);
15    }
16    i++; // Don't forget to increment the index!
17 }
```

Submitted

Test Input

4

Output

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QUESTION 12:

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10

Your Fixed Code

```
1 int dequeue() {
2     if (front == -1) return -1; // empty queue
3
4     int data = queue[front];
5
6     // move front forward
7     if (front == rear) { // only one element was present
8         front = -1;
9         rear = -1;
10    } else {
11        front = (front + 1) % MAX;
12    }
13
14    return data;
15 }
```

Submitted

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192525224
RB_6a95a0f732a491_64260419\main.c
: In function 'dequeue':
C:\Windows\TEMP\sandbox_192525224
RB_6a95a0f732a491_64260419\main.c

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QUESTION 13:



SIMATS Online Programming Lab

DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

```
1 int binarySearch(int arr[], int l, int r, int x) {
2     if (l > r) return -1;
3
4     int mid = l + (r - l) / 2;
5
6     if (arr[mid] == x) return mid;
7
8     if (arr[mid] > x)
9         return binarySearch(arr, l, mid - 1, x);
10    else
11        return binarySearch(arr, mid + 1, r, x);
12 }
```

Submitted

6

Output

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QUESTION 14:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10

Your Fixed Code

```
1 void insertFront(int data) {
2     // Assuming front and rear indices start at -1 when deque is empty
3     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
4         // Deque is full
5         return;
6     }
7
8     if (front == -1) {
9         // First insertion
10        front = rear = 0;
11    } else if (front == 0) {
12        front = MAX - 1;
13    } else {
14        front = front - 1;
15    }
16
17    deque[front] = data;
18 }
```

Submitted

Test Input

5

Output

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QUESTION 15:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Your Fixed Code

```
1 void enqueue(int data) {
2     if ((rear + 1) % MAX == front) return; // full condition
3     queue[rear] = data;
4     rear = (rear + 1) % MAX;
5 }
```

Submitted

Test Input

stdin input (optional)

Output

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QUESTION 16:



SIMATS Online Programming Lab

DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

Your Fixed Code

```
1 void deleteNode(Node* del) {
2     // Check if the node to be deleted is NULL
3     if (del == NULL) {
4         return;
5     }
6
7     // If the node to be deleted is NOT the head node
8     if (del->prev != NULL) {
9         del->prev->next = del->next;
10    }
11
12    // If the node to be deleted is NOT the last node
13    if (del->next != NULL) {
14        del->next->prev = del->prev;
15    }
16
17    // Free the memory occupied by the node
18    delete del;
19 }
```

Submitted

Test Input

stdin input (optional)

Output

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QUESTION 17:



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DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Your Fixed Code

```
1 int POP() {
2     if (top == 0) { // underflow: stack is empty
3         printf("Underflow\n");
4         return -1;
5     }
6     top--; // remove top element
7     return stack[top];
8 }
```

Test Input
stdin input (optional)

Output

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Submitted

QUESTION 18:



SIMATS Online Programming Lab
DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int x) {
9     if (top == MAX - 1) {
10         printf("Overflow\n");
11         return;
12     }
13     top++;
14     stack[top] = x;
15 }
16
17 void pop() {
18     if (top == -1) {
19         printf("Underflow\n");
20         return;
21     }
22     printf("Popped: %d\n", stack[top]);
23     top--;
24 }
25
26 void display() {
27     if (top == -1) {
```

Test Input
stdin input (optional)

Output
Compilation Error:
C:\Windows\TEMP\sandbox_192525224
RB_6a95a1e51aab69.31184086\main.c
:1:1: error: expected identifier
or '(' before 'if'
1 | if top == MAX:

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QUESTION 19:



SIMATS Online Programming Lab
DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Your Fixed Code

```
1 // Assuming 'front' and 'rear' are initialized to -1
2 // and 'MAX' is the size of the queue.
3
4 if ((rear + 1) % MAX == front) {
5     printf("Queue Full\n");
6 } else {
7     if (front == -1) front = 0; // First element being added
8     rear = (rear + 1) % MAX;
9     queue[rear] = vehicle;
10 }
```

Submitted

Test Input

stdin input (optional)

Output

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QUESTION 20:



SIMATS Online Programming Lab
DS PROGRAMMING

BHAVANASI SATHVIKA
192525224RB

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def PEEK(stack, top):
2     if top == -1:
3         return "Stack is empty"
4     return stack[top]
```

Submitted

Test Input

stdin input (optional)

Output

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